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Occurrence of *bla*_{OXA-116} Carbapenemase in *Escherichia coli* ST2519 of Clinical Origin: A Report from Northeast India

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Carbapenem-resistant *Escherichia coli* pose a significant threat to global public health due to the dearth of available treatment options, resulting in infections with high mortality and morbidity. The study aimed to investigate the mechanism of carbapenem resistance in a carbapenem non-susceptible *E. coli* isolate recovered from an urinary tract infection patient admitted to a tertiary referral hospital, through whole-genome sequencing using Illumina NovaSeq 6000 platform. Carbapenemase production followed by antibiotic susceptibility testing were performed following Clinical Laboratory Standard Institute guidelines. Polymerase chain reaction targeting carbapenemase genes was performed followed by an investigation of horizontal transferability. The Center for Genomic Epidemiology database was used to analyze the sequenced data. ST2519 *E. coli* BJD_EC1808 with a genome size of 5.8 Mb harbored Col440I plasmid and a chromosomally located *bla*_{OXA-116} gene with an IS18 element upstream, along with multiple antibiotic resistance genes conferring clinical resistance toward beta-lactams, aminoglycosides, amphenicols, sulfonamides, tetracyclines, trimethoprim, rifampin, macrolide, and streptogramin antibiotics and antiseptics. *E. coli* ST2519 harboring *bla*_{OXA-116} associated with a mobile genetic element exhibiting carbapenem resistance is a public health threat due to its limiting effect on the therapeutic usage of carbapenem and their dissemination into carbapenem non-susceptible phenotypes will contribute to carbapenem resistance burden and, therefore, warrants urgent monitoring and clinical intervention.

Keywords: antimicrobial resistance, carbapenem resistance, carbapenem-resistant *Escherichia coli*, carbapenemase, *bla*_{OXA-116}, whole-genome sequencing

Introduction

Carbapenem-resistant *Escherichia coli* (CREco) a notorious pathogen prevailing in the health sector, has been categorized by the World Health Organization (WHO) as pathogen of critical priority. This was because CREco is associated with numerous clinical manifestations that have high morbidity and mortality rates and other factors such as their rate of transmission, multidrug resistance, and dearth of available treatment options, thereby posing an imminent threat to global public health.¹ The usage of carbapenems, considered “antimicrobials of last resort” for treating infections caused by multidrug

resistant gram-negative bacteria, is thwarted by the emergence of carbapenem-resistant bacteria primarily due to the production of carbapenemase and acquisition of carbapenem-resistance determinants.² Carbapenemase-encoding genes are often found associated with mobile genetic elements that aid in conferring high level of clinical resistance to carbapenems along with their intercellular and intracellular dissemination, thereby increasing antibiotic resistance burden which is at present a global concern due to its limiting effect on therapeutic options. Both broad- as well as narrow-host range plasmids were reported to be associated with the carriage of carbapenemase-encoding genes. In *E. coli*, plasmids of diverse

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incompatibility types such as Inc A/C, Inc F_{REP}B, Inc K/B, Inc N, Inc L/M, and Inc X were identified with the transmission of different classes of carbapenemase.^{2–4} Since its first report in 1982, different carbapenemases and their variants with their versatile hydrolytic abilities have put forward new challenges to clinicians worldwide in their management.²

The first report of OXA-51 was from Argentina in 2005 within isolates of *Acinetobacter baumannii*.⁵ The gene was first thought intrinsic to *Acinetobacter* species; however, over time, this *bla*_{OXA-51} gene and its variants were also reported from Enterobacterales and *Pseudomonas*.^{6,7} OXA-116, a variant of *bla*_{OXA-51}, was first reported in 2010 within a clinical strain of carbapenem-resistant *A. baumannii* from Shenzhen, China.³ The affinity of *bla*_{OXA-116} against carbapenems is low and would not confer carbapenem resistance, although exhibiting hydrolytic activity against imipenem and meropenem.⁴ However, a strong transcriptional promoter in the upstream region of the gene associated with mobile genetic elements increases the expression of this gene and can contribute to carbapenem resistance, thereby compromising therapeutic options.^{8–10} Studies indicate that under exposure to selective carbapenem pressure, these mobile genetic elements also contribute to the maintenance and expression of carbapenemase genes within the bacterial host.^{2,11–13}

CREco has also been labeled in the red category of pathogens in the Indian Priority Pathogen List that was jointly developed by the Department of Biotechnology, Government of India (DBT, GoI) and the WHO (India office).¹⁴ Therefore, with this background and emphasizing on the current scenario of carbapenem resistance worldwide and in India attributed to carbapenemase-encoding genes, this study reports a *bla*_{OXA-116}-harboring CREco (BJD_EC1808) isolated from the urine sample of a patient of a tertiary referral hospital in the northeastern part of India.

Materials and Methods

Study setting and bacterial isolate

This research work was carried out in the Department of Microbiology, Assam University, Silchar. In December 2019, a carbapenem non-susceptible *E. coli* (BJD_EC1808), identified using the VITEK® 2 Compact Automated System (bioMérieux, France), was received from the Department of Microbiology, Silchar Medical College and Hospital, Silchar, Assam, India. The isolate was recovered from the urine sample of a catheterized male patient admitted to the surgery ward of the tertiary referral hospital.

Screening of carbapenemase production

Carbapenemase production in the study isolate was phenotypically investigated using the HiCrome™ KPC Agar (HiMedia, India) supplemented with HiCrome™ KPC Selective Supplement (HiMedia, India) and Rapidec® Carba NP test (bioMérieux, France) as per the manufacturer's instructions using *E. coli* ATCC 25922 as a negative control.

Antibiotic susceptibility testing

The Kirby–Bauer disc diffusion method was performed to investigate the susceptibility of the study isolate against the following antibiotics: ampicillin (10 µg), cefepime (30 µg),

ceftriaxone (30 µg), cefotaxime (30 µg), ceftazidime (30 µg), aztreonam (30 µg), ertapenem (10 µg), imipenem (10 µg), meropenem (10 µg), amikacin (10 µg), gentamicin (10 µg), and ciprofloxacin (5 µg) (HiMedia, India). The minimum inhibitory concentrations (MICs) of ertapenem (MSD, France), imipenem (Merck, France), and meropenem (AstraZeneca, UK) were also examined using the agar dilution method (concentration range: 1–256 µg/mL). The tests were performed following the Clinical Laboratory Standard Institute (CLSI) 2022 guidelines using *E. coli* ATCC 25922 as a quality control strain.¹⁵

Molecular detection of carbapenemase

Total DNA was extracted from the study isolate using the boiling centrifugation method and investigated for carbapenemase genes through the polymerase chain reaction (PCR) assay. PCR was performed using primer pairs and reaction conditions described earlier (Supplementary Table S1) and the amplified product was sequenced for confirmation of carbapenem-resistance gene.^{16–19}

Horizontal gene transferability of *bla*_{OXA-116}

The transferability of *bla*_{OXA-116} was investigated through transformation and conjugation assays. Plasmid was extracted using the QIAprep Spin Miniprep Kit (Qiagen, Germany) as per the manufacturer's instructions. Transformation assay was performed by the heat-shock method using *E. coli* DH5α as a recipient strain. The *bla*_{OXA-116} transformants were selected on Luria Bertani (LB) agar (HiMedia, India) supplemented with 0.5 µg/mL of imipenem (Merck, France).²⁰ To assess the self-transferability of *bla*_{OXA-116}, a conjugation experiment was performed using a sodium azide-resistant *E. coli* J53 as the recipient strain. The donor cell harboring *bla*_{OXA-116} and the recipient cell were inoculated in LB broth (HiMedia, India) and incubated at 37°C till optical density reached 0.8–0.9 at A₆₀₀. Subsequently, mating was done by culturing both the donor and the recipient cells together at a ratio of 1:5; 100 µL of the cell mixture was then cultured on LB agar (HiMedia, India) medium supplemented with 0.5 µg/mL of imipenem (Merck, France) and 100 µg/mL of sodium azide (HiMedia, India) and incubated at 37°C overnight for selection of transconjugants.²¹

Cloning of *bla*_{OXA-116} and antibiogram of clones

The whole gene of *bla*_{OXA-116} was amplified through PCR assay using the amplification conditions of 95°C for 2 minutes followed by 35 cycles of 95°C for 45 seconds, 47°C for 45 seconds, and 72°C for 1.20 minutes with a single final extension cycle of 72°C for 5 minutes. The primer pair (OXA-116_WF: GAAGACCACGTCCTTGCACT and OXA-116_WR: GATCAAGGTCGCGGAAAAGG) was designed using the NCBI primer blast tool (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>) from the flanking regions of *bla*_{OXA-116} gene to amplify the whole gene including its native promoter. A volume of 50 µL reaction mixture was prepared for the PCR assay, which included 25 µL of 2× GoTaq® Green Master Mix (Promega, Madison, USA), 1 µL of each primer (10 pmol/µL), 2 µL of DNA template (~100 ng/µL), and nuclease-free water. The amplified PCR product (1037 bp) was then purified using MinElute® PCR Purification Kit (Qiagen, Germany) and ligated into pMD20-T vector as per manufacturer's protocol

(Mighty TA-cloning Kit, TaKaRa, Japan). The recombinant plasmid was then transformed into *E. coli* DH5 α by the heat-shock method and the clones were selected by blue-white screening on LB agar medium (HiMedia, India) containing ampicillin (100 μ g/mL) and imipenem (0.5 μ g/mL) and were further confirmed by colony PCR. Kirby–Bauer disc diffusion method and broth microdilution method were performed as per CLSI 2022 guidelines to investigate the susceptibility of the clones against ertapenem, imipenem, and meropenem.¹⁵

Whole-genome sequencing and assembly

Whole-genome sequencing (WGS) was carried out using the Illumina NovaSeq 6000 platform (outsourced to Bionivid Technology Private Limited, Bengaluru, India). Quality control and data filtering were performed using Fastp (version 0.20.0) with standard parameters.²² *De novo* assembly and scaffolding after quality trimming of the reads were conducted using SPAdes (version 3.13.0).²³ The 16S rRNA gene sequence was predicted using Metaerg tool (version 1.2.0) and the nearest genome reference was identified using NCBI BLAST® tool (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>). Genomes were oriented and rearranged using web-based tool MeDuSa (version 2.2) using default web-interface parameters.²⁴ Genomes were annotated using Prokka software (version 1.11.1).²⁵

Bacterial species of the study isolate was further confirmed through SpeciesFinder 2.0 (<https://cge.food.dtu.dk/services/SpeciesFinder/>), whereas its sequence type, phylotype, and serotype were identified using MLST 2.0 (version 2.0.9, <https://cge.food.dtu.dk/services/MLST/>), In Silico Clermont Phylotyper (<https://ezclermont.hutton.ac.uk/>), and SerotypeFinder 2.0 (version 2.0.1, <https://cge.food.dtu.dk/services/SerotypeFinder/>), respectively. Antimicrobial resistance genes carried by the isolate were identified using ResFinder (version 4.4.2, <http://genepi.food.dtu.dk/resfinder>). In addition, plasmids and other mobile genetic elements and their relation to antibiotic resistance determinants carried by the isolate were identified through PlasmidFinder 2.1 (version 2.0.1, <https://cge.food.dtu.dk/services/PlasmidFinder/>) and MobileElementFinder (version v1.0.3, <https://cge.food.dtu.dk/services/MobileElementFinder/>), respectively. Virulence gene acquired by the study isolate was detected via VirulenceFinder 2.0 (version 2.0.3, <https://cge.food.dtu.dk/services/VirulenceFinder/>), whereas PathogenFinder 1.1 (version 1.1, <https://cge.food.dtu.dk/services/PathogenFinder/>) was used to predict the potential pathogenicity of BJD_EC1808.

Results

E. coli (BJD_EC1808) was isolated on 24.12.2019 from the urine sample of a male patient with a symptom of urinary tract infection (UTI) from the surgery ward of Silchar Medical College and Hospital, Silchar, India. The isolate exhibited resistance toward all the antibiotics used in susceptibility testing, as BJD_EC1808 harbored multiple antibiotic resistance genes in its resistome conferring resistance to extended-spectrum β -lactams (*bla*_{TEM-116}), aminoglycosides (*aadA1*, *ant*(2'')-Ia, *aph*(3')-IIa, *aph*(6)-Id, *ant*(2'')-IIc, *ant*(3'')-IIa, and *rsmA*), amphenicols (*cmlA1*, *catA1*), sulfonamides (*sul1*, *sul2*), trimethoprim (*dhfrA1*), tetracycline (*tetB*), and rifampin (*ARR-2*) (Table 1, Fig. 1). The MIC of carbapenems for this carbapenemase-producing isolate was above resistance breakpoint (≥ 64 μ g/mL). The isolate

harbored a variant of *bla*_{OXA-51} gene, i.e., *bla*_{OXA-116}, for which the transferability attempts through transformation and conjugation assays were unsuccessful. The recombinant plasmid containing the whole gene of *bla*_{OXA-116} in pMD20-T vector was successfully transformed into *E. coli* DH5 α . The clones of *bla*_{OXA-116} screened via blue-white screening were confirmed through colony PCR. The *bla*_{OXA-116} clones were resistant to carbapenems, and the MIC value of carbapenems was above breakpoint, ertapenem (16 μ g/mL), imipenem (16 μ g/mL), and meropenem (32 μ g/mL). WGS data revealed that the *bla*_{OXA-116} differs from its parent *bla*_{OXA-51} by 39 base substitutions that resulted in seven amino acid changes (Val to Ala at position 19, Asp88 to Asn, Lys117 to Asn, Pro165 to Gln, Lys166 to Glu, Met174 to Ile, and Asp to Asn at position 196). The *bla*_{OXA-116} gene was flanked by IS18 and *aadA1* in the upstream region while *rsmA* and *ant*(2'')-Ia were in the downstream area of this chromosomally located gene (Fig. 2). It was observed that the addition of IS18 upstream of *bla*_{OXA-116} resulted in the generation of a potential promoter sequence consisting of –35 and –10 elements, which were separated by 17 nucleotides, a distance considered optimum for promoter function. The –35 (TTGCCG) motif was located within the insertion sequence, whereas the –10 (TAAAAT) motif was found adjacent to the site of insertion. In addition, two multidrug efflux genes *msrE* and *qacE* mediating resistance toward macrolide and streptogramin antibiotics and antiseptics, respectively, were also identified in BJD_EC1808. The MLST scheme (Achtman) of BJD_EC1808, which was based on the sequences of seven housekeeping genes (*adh*, *fumC*, *gyrB*, *icd*, *mdh*, *purA*, and *recA*), confirmed that BJD_EC1808 belonged to *E. coli* sequence type ST2519. The serotype was determined by the genotype of *fliC*, *wzx*, and *wzy*, and the results showed that BJD_EC1808 belonged to the O102-H6 serotype. BJDEC_1808 was a group B2 *E. coli* and harbored Col440I plasmid replicon type. The pathogenicity of BJD_EC1808 was predicted by comparing its protein to a protein family database associated with pathogenic and non-pathogenic bacteria. The proteins in BJD_EC1808 matched with 201 pathogenic families and it was predicted to be a human pathogen (probability of 0.85). Virulome of BJD_EC1808 includes a number of genes that encodes for several virulence factors such as adhesins (*afaA*, *fimH*, *papC*), toxins (*hlyA*, *senB*), iron uptake (*fyuA*, *iucC*, *iutA*), and capsule (*kpsE*, *kpsMII*), which are summarized along with their functions in Table 1. The nucleotide sequences of *bla*_{OXA-116} has been deposited in GenBank under the accession numbers OR188143.

Discussion

CREco is one of the main pathogens of carbapenem-resistant Enterobacterales (CRE) that causes a wide range of clinical infections that are associated with high morbidity and mortality rates and their upsurge have threatened the usage of carbapenems, broad-spectrum β -lactam antibiotics that are considered drugs of last resort, against multidrug-resistant gram-negative bacteria infections, thereby posing a serious threat to global public health.

The surge in reports of CRE in India, over the years, and their limiting effect on prevailing therapeutic options and other associated factors, have resulted in their categorization as pathogens of “critical priority” in the Indian Priority Pathogen List, jointly published by the DBT, GoI, and the WHO.^{7,26–29} The present work

TABLE 1. ANTIMICROBIAL GENE PROFILE OF BJD_EC1808 CO-HARBORING *BLA*_{OXA-116}

Isolates ID	Susceptibility profile				β-Lactamases genes	Aminoglycoside resistance genes	Other resistance genes	Mobile genetic elements	Sequence type	Virulence factors
	Antibiotics	MIC value								
		Parent strain	Clone							
BJD_EC1808	Ampicillin	≥256 µg/mL	—	<i>bla</i> _{OXA-116} , <i>bla</i> _{TEM-116}	<i>aadA1</i> , <i>ant</i> (2'')-Ia, <i>aph</i> (3')-IIa, <i>aph</i> (6)-Id, <i>ant</i> (2'')-IIc, <i>ant</i> (3'')-IIa, <i>rsmA</i>	<i>catA1</i> , <i>cmIA1</i> , <i>sul1</i> , <i>sul2</i> , <i>dfrA1</i> , <i>tet</i> (B), <i>ARR-2</i> , <i>msr</i> (E), <i>qacE</i>	Col440I, IS18	ST2519	Adhesins <i>afaA</i> : transcriptional regulator <i>afaD</i> : afimbrial adhesion <i>fimH</i> : Type 1 fimbriae <i>iha</i> : adhesion protein <i>papA_F7-2</i> : major pilin subunit F7-2 <i>papC</i> : outer membrane usher P fimbriae Toxins <i>hlyA</i> : α-hemolysin <i>iss</i> : increased serum survival <i>senB</i> : plasmid encoded enterotoxin Iron uptake <i>chuA</i> : outer membrane hemin receptor <i>fyuA</i> : siderophore receptor <i>irp2</i> : non-ribosomal protein synthetase <i>iucC</i> : aerobactin synthetase <i>iutA</i> : ferric aerobactin receptor Capsule <i>kpsE</i> : capsule polysaccharide export inner-membrane protein <i>kpsMII</i> : polysialic acid transport protein, group 2 capsule	
	Aztreonam	≥256 µg/mL	—							
	Amikacin	≥256 µg/mL	—							
	Cefepime	≥256 µg/mL	—							
	Ceftazidime	≥256 µg/mL	—							
	Ceftriaxone	≥256 µg/mL	—							
	Cefotaxime	≥256 µg/mL	—							
	Ciprofloxacin	≥256 µg/mL	16 µg/mL							
	Ertapenem	≥64 µg/mL	—							
	Gentamicin	≥256 µg/mL	16 µg/mL							
	Imipenem	≥64 µg/mL	32 µg/mL							
	Meropenem	≥128 µg/mL	—							

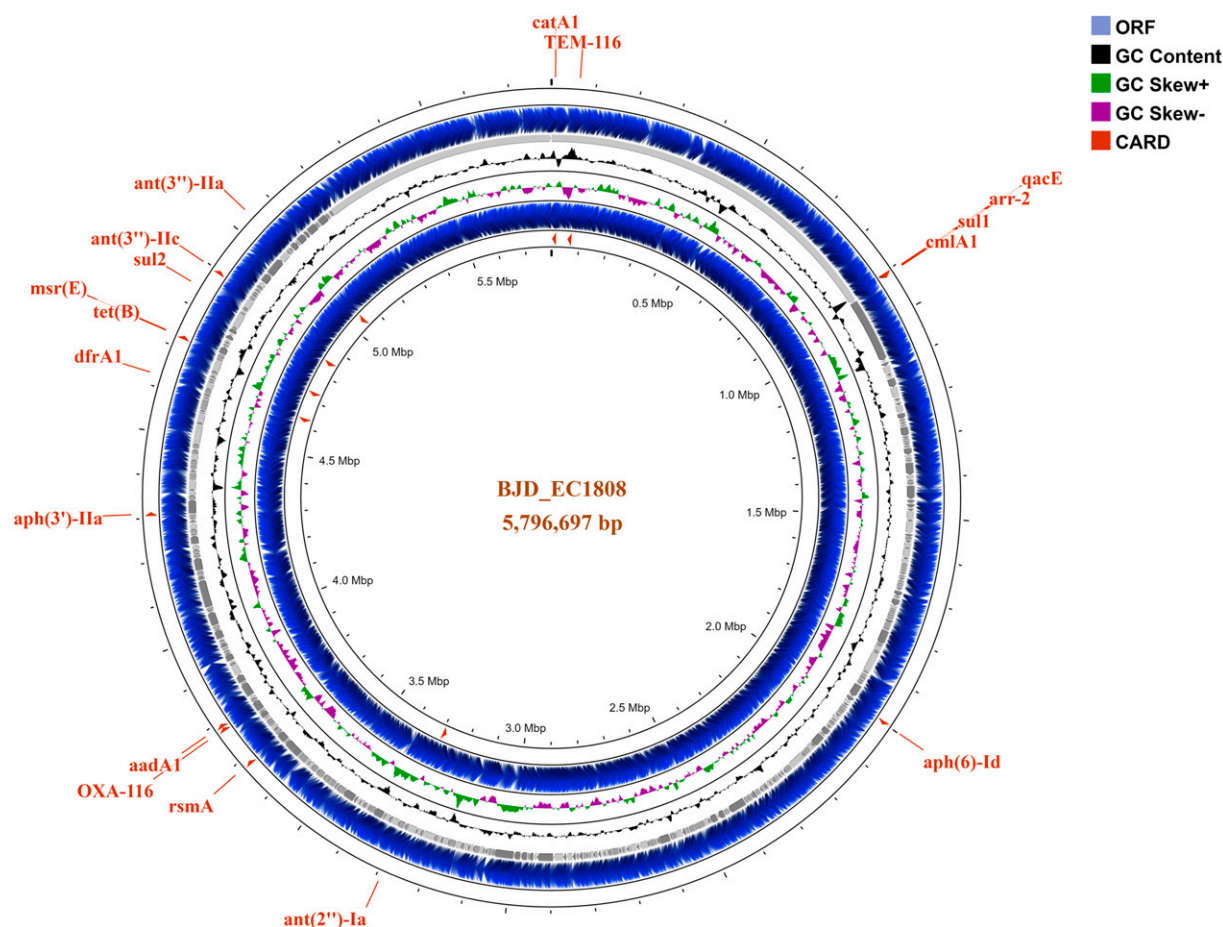


FIG. 1. Circular genome map of *E. coli* BJD_EC1808. The scale indicates the location in Mbp (chromosome), starting with the initial coding region. The inner and outermost circles represent the backward and forward strands illustrating the coding sequences. The second and third circles show the GC skew and GC content, respectively.

reports a CREco harboring *bla*_{OXA-116}, a variant of *bla*_{OXA-51} isolated from the urine of an UTI patient from a tertiary referral hospital in the northeastern part of India. OXA-51-like genes were first thought intrinsic to *A. baumannii* and were being used as identification markers for this species; however, lately, this gene has been reported in several studies within Enterobacterales and other non-lactose fermenting gram-negative bacteria, highlighting their interspecific dissemination. A recent study conducted in 2022 from Iran, reported the carriage of the *bla*_{OXA-51-like} gene in Enterobacterales isolates recovered from urine samples of UTI patients from a hospital of Tehran, Iran.³⁰ Similarly, Ibrahim and his team in 2022, in their cross-sectional hospital-based study, reported the presence of *bla*_{OXA-51} gene in *E. coli* isolated from urosepsis patients.⁴ In 2013, variants of *bla*_{OXA-51} gene were identified within Enterobacterales isolates obtained from Mercy Hospital, Bo, Sierra Leone, by Leski *et al.*³¹ The findings of the above research work suggest the interspecific dissemination of the *bla*_{OXA-51} gene and its variants, which is also in congruence with our study findings. However, to the best of our knowledge, this is the first report of the occurrence of *bla*_{OXA-116} gene within CREco of clinical origin from India.

Insertion sequence elements provide a strong outward promoter that aid in better expression of otherwise silent *bla*_{OXA} genes encoding carbapenemases, thereby conferring clinical resistance to carbapenems.^{4,32–35} The insertion sequence

IS18, a member of IS30 insertion sequence family, was identified upstream of *bla*_{OXA-116} gene that might have played a role in the study isolate BJD_EC1808 exhibiting high MICs against carbapenems. In addition, mobile genetic elements associated with carbapenem-resistance determinants pose a serious health hazard as they can also serve as a genetic vehicle for the dissemination of resistance genes into carbapenem-susceptible phenotypes, thereby increasing antibiotic resistance burden, which is at present a significant global concern. The study also highlights the positive selection pressure generated by the surge in usage of carbapenems within the study center that might have aided in the evolution, maintenance, and expression of this *bla*_{OXA-51} variant, *bla*_{OXA-116} gene by mobile genetic element within *E. coli* ST2519, thereby conferring clinical resistance to carbapenems, antibiotics of last resort and limiting our therapeutic options. It was opined that isolates harboring carbapenemase-encoding genes often carry additional resistance genes that confer resistance to other β -lactams, aminoglycosides, fluoroquinolones, sulfonamides, tetracyclines, and other antibiotics.^{36–38} In accordance, our study isolates also co-harbored multiple resistance genes elucidating their multidrug-resistant nature, correlating with the observations of antibiotic susceptibility testing. Although how these resistance mechanisms work with each other needs further investigation, these findings suggest they work in combination to confer enhanced resistance against carbapenems as well as

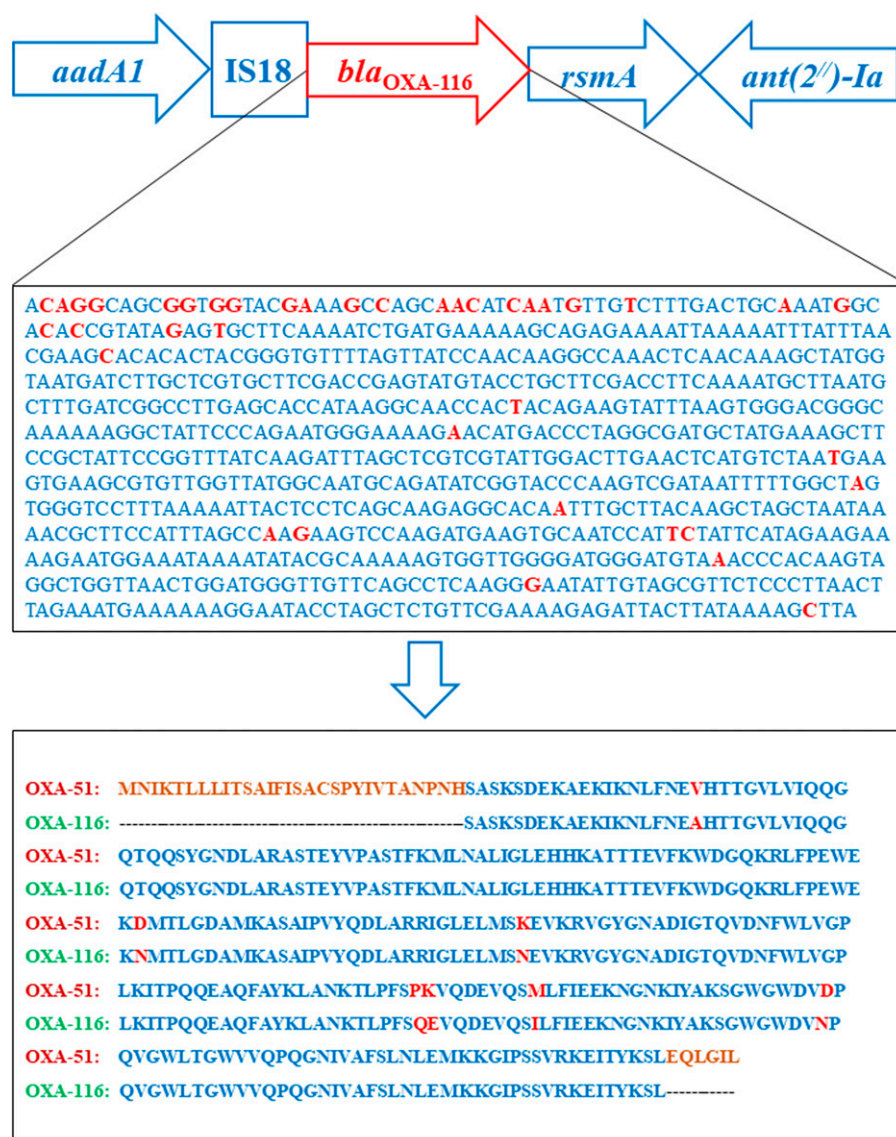


FIG. 2. Schematic representation of the genetic context of *bla*_{OXA-116} gene that differs from its parent *bla*_{OXA-51} by 39 base substitutions (red color) that resulted in seven amino acid changes (red color): Val19 to Ala, Asp88 to Asn, Lys117 to Asn, Pro165 to Gln, Lys166 to Glu, Met174 to Ile, and Asp to Asn at position 196.

other antibiotics of preferred treatment regime, thus limiting our antimicrobial resources.

Considering the fluid nature of carbapenemase genes, *bla*_{OXA-116} gene aided by mobile genetic elements conferring clinical resistance toward carbapenems poses a serious health hazard. As heavy carbapenem use in hospitals, especially in developing countries, is already an established risk factor for emergence of carbapenem-resistant organisms, our findings warrant urgent monitoring of these potential sources and vehicle of future dissemination of resistance as they pose a serious threat to the control of antimicrobial resistance and endanger our fight against it.

Conclusions

E. coli ST2519 harboring the carbapenemase gene *bla*_{OXA-116} and exhibiting resistance toward carbapenems raises a significant concern on the use of this antibiotic of last resort as a therapeutic

option. With an increase in reports of class D carbapenemases and its variants among clinically significant gram-negative bacteria, and considering the fluid nature of carbapenem-resistance determinants aided by mobile genetic elements, the present study is of epidemiological importance and warrants for surveillance of this IS18-associated carbapenem-resistance determinants along with the development of screening methods for their routine detection to optimize clinical intervention in order to avoid treatment failure. Further investigations are required to understand the inducing factors responsible for the intra- and interspecific dissemination or expression of this carbapenem-resistance determinant with an aim to thwart further spread of carbapenem resistance in this region and worldwide.

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Authors' Contributions

B.J.D.: Investigation, formal analysis, data curation, and writing—original draft. K.M.S.: Resources and formal analysis. J.W.: Formal analysis. D.D.C.: Methodology, validation, and writing—review and editing. A.B.: Conceptualization and supervision. All authors ensured that this is the case. All the authors read and approved the final article.

Data Availability Statement

The nucleotide sequence of *bla*_{OXA-116} generated and/or analyzed during the current study is available in GenBank repository under the accession number OR188143.

Authors' Disclosure Statement

The authors declare that the study was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary Material

Supplementary Table S1

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